

G11 T1 C6 – Experimental Interval Probabilities

Suppose a sample contains n observations and exactly k of them satisfy the stated condition. The experimental probability, or relative frequency, is

$$\hat{P}(a \leq X \leq b) = \frac{k}{n},$$

where n is the sample size and k is the number of observations for which $a \leq X \leq b$. It may be reported in three equivalent forms:

$$\hat{P} = \frac{k}{n} = \text{decimal form} = (100 \times \text{decimal form})\%.$$

The same counting method applies to conditions using $<$, $\leq$, $>$, or $\geq$. Pay close attention to whether each endpoint is included.

1. Water-bottle filling lines (multiple samples).

Three filling lines produced the following sample fill volumes, in mL:

Line A	496	500	503	499	505	501	497	502
Line B	494	498	500	506	501	503	499	502
Line C	497	501	504	500	498	502	503	499

For the combined sample from all three lines, estimate $\hat{P}(499 \leq X \leq 502)$. Give the answer as a fraction, a decimal, and a percentage.

2. Bike commute times (raw data).

The commute times, in minutes, for 15 rides were

18, 22, 25, 19, 31, 24, 27, 20,
23, 29, 21, 26, 17, 24, 28.

Estimate $\hat{P}(X < 24)$ as a fraction, a decimal, and a percentage.

3. Books borrowed (value–frequency table).

A librarian recorded the number of books borrowed by 40 visitors.

Books x	0	1	2	3	4	5	6
Frequency f	3	7	10	9	6	4	1

Estimate $\hat{P}(X \geq 3)$ as a fraction, a decimal, and a percentage.

4. Parcel masses (class-interval table).

The masses of 50 parcels are grouped below. Each class includes its left endpoint but not its right endpoint.

Mass x (kg)	[0, 2)	[2, 4)	[4, 6)	[6, 8)	[8, 10)
Frequency	4	11	17	12	6

Estimate $\hat{P}(2 \leq X < 8)$ as a fraction, a decimal, and a percentage.

5. Seedling growth (raw data).

The increases in height, in cm, of 20 seedlings were

4.2, 5.0, 5.7, 6.1, 4.8, 5.5, 6.0, 6.4, 5.2, 4.5,
5.9, 6.3, 5.0, 5.4, 4.9, 6.0, 5.8, 6.6, 5.1, 5.6.

Estimate $\hat{P}(5.0 < X \leq 6.0)$ as a fraction, a decimal, and a percentage.

6. Customer service calls (value–frequency table).

A service center summarized call lengths, rounded to the nearest minute.

Length x (min)	2	3	4	5	6	7	8	9
Frequency f	2	5	8	11	9	7	5	3

Estimate both (a) $\hat{P}(X > 6)$ and (b) $\hat{P}(4 \leq X < 8)$. Give each answer in all three forms.

7. Daily solar-energy output (class-interval table).

The output recorded on 80 days is grouped below. Each class includes its left endpoint but not its right endpoint.

x (kWh)	[0, 5)	[5, 10)	[10, 15)
Freq.	5	12	21
x (kWh)	[15, 20)	[20, 25)	[25, 30)
Freq.	24	13	5

Estimate both (a) $\hat{P}(X \geq 15)$ and (b) $\hat{P}(10 \leq X < 25)$. Give each answer in all three forms.

1 Solutions

C6.1 – Water-bottle filling lines

There are 8 observations from each of three lines, so

$$n = 3(8) = 24.$$

The qualifying values are 500, 499, 501, 502 from line A; 500, 501, 499, 502 from line B; and 501, 500, 502, 499 from line C. Thus,

$$k = 4 + 4 + 4 = 12.$$

Therefore,

$$\hat{P}(499 \leq X \leq 502) = \frac{k}{n} = \frac{12}{24} = \frac{1}{2} = 0.50 = 50\%.$$

C6.2 – Bike commute times

There are $n = 15$ rides. The times below 24 are 18, 22, 19, 20, 23, 21, 17, so $k = 7$. Therefore,

$$\hat{P}(X < 24) = \frac{k}{n} = \frac{7}{15} \approx 0.467 \approx 46.7\%.$$

C6.3 – Books borrowed

The sample size is the sum of all frequencies:

$$n = 3 + 7 + 10 + 9 + 6 + 4 + 1 = 40.$$

For $X \geq 3$, sum the frequencies at 3, 4, 5, and 6:

$$k = 9 + 6 + 4 + 1 = 20.$$

Therefore,

$$\hat{P}(X \geq 3) = \frac{k}{n} = \frac{20}{40} = \frac{1}{2} = 0.50 = 50\%.$$

C6.4 – Parcel masses

The sample size is

$$n = 4 + 11 + 17 + 12 + 6 = 50.$$

The included buckets are $[2, 4)$, $[4, 6)$, and $[6, 8)$. Hence,

$$k = 11 + 17 + 12 = 40.$$

Therefore,

$$\hat{P}(2 \leq X < 8) = \frac{k}{n} = \frac{40}{50} = \frac{4}{5} = 0.80 = 80\%.$$

C6.5 – Seedling growth

There are $n = 20$ seedlings. The values satisfying $5.0 < X \leq 6.0$ are

$$5.7, 5.5, 6.0, 5.2, 5.9, 5.4, 6.0, 5.8, 5.1, 5.6,$$

so $k = 10$. Notice that the two values equal to 5.0 are excluded, while the two values equal to 6.0 are included. Therefore,

$$\hat{P}(5.0 < X \leq 6.0) = \frac{k}{n} = \frac{10}{20} = \frac{1}{2} = 0.50 = 50\%.$$

C6.6 – Customer service calls

For both parts, the sample size is

$$n = 2 + 5 + 8 + 11 + 9 + 7 + 5 + 3 = 50.$$

(a) For $X > 6$, sum the frequencies at 7, 8, and 9:

$$k = 7 + 5 + 3 = 15.$$

Thus,

$$\hat{P}(X > 6) = \frac{k}{n} = \frac{15}{50} = \frac{3}{10} = 0.30 = 30\%.$$

(b) For $4 \leq X < 8$, sum the frequencies at 4, 5, 6, and 7:

$$k = 8 + 11 + 9 + 7 = 35.$$

Thus,

$$\hat{P}(4 \leq X < 8) = \frac{k}{n} = \frac{35}{50} = \frac{7}{10} = 0.70 = 70\%.$$

C6.7 – Daily solar-energy output

For both parts, the sample size is

$$n = 5 + 12 + 21 + 24 + 13 + 5 = 80.$$

(a) The included buckets are $[15, 20)$, $[20, 25)$, and $[25, 30)$.

Therefore,

$$k = 24 + 13 + 5 = 42,$$

and

$$\hat{P}(X \geq 15) = \frac{k}{n} = \frac{42}{80} = \frac{21}{40} = 0.525 = 52.5\%.$$

(b) The included buckets are $[10, 15)$, $[15, 20)$, and $[20, 25)$.

Therefore,

$$k = 21 + 24 + 13 = 58,$$

and

$$\hat{P}(10 \leq X < 25) = \frac{k}{n} = \frac{58}{80} = \frac{29}{40} = 0.725 = 72.5\%.$$